

MUHAMMED MUIZ ARUMMAL

Birmingham, UK | abdulmuiz3570@gmail.com

secslayer.github.io | github.com/secslayer | linkedin.com/in/muhammed-muiz-111ba3362

PROFILE

MSc Artificial Intelligence & Machine Learning, University of Birmingham (2026–27), based in the UK. AI-security engineer red-teaming and evaluating LLM and agentic systems: benchmarked 3 attacker architectures against a live multi-agent target and reproduced a 2026 jailbreak paper across 3 models. Defensive base in SOC telemetry (a pipeline normalising 45K+ security events) and detection engineering (CompTIA CySA+). Current focus: the reliability of the automated judges the jailbreak field trusts to score attacks.

RESEARCH EXPERIENCE

Judge-Shaped, Not Harm-Shaped

preprint in preparation | github.com/secslayer/jlens-fuzz

- Found the standard **hubert233/GPTFuzz** judge — reused across the jailbreak literature without re-validation — inflates attack success by rewarding jailbreak-*shaped* persona/roleplay framing over actual harmful content.
- Built a stricter two-stage replacement (deterministic refusal pre-filter + anti-roleplay rubric judge); reported honestly it *still* passed a completion that refuses and offers crisis-support resources — the failure survives a targeted fix. Smoke-scale (n=5/condition) under a hard free-tier budget, stated plainly.

AJF reproduction

github.com/secslayer/reproductions

- Independent end-to-end reproduction of a 2026 adaptive-jailbreak framework (arXiv:2505.23404) — prompt mutation, binary-tree encoding, dual-cipher output encryption, capability probe — across three local models, plus an ablation and the adaptive defense the paper never evaluates.
- Replicated the judge-reliability finding (a self-judge wrong in *both* directions on the same prompts; all verdicts re-checked by hand); also reproduced **GPTFuzzer** end to end.

Agent-protocol authorization audit

deployed multi-agent (A2A) platform | security audit & fix

- Audited authorization on a deployed multi-agent (A2A) platform against the 14-row protocol threat matrix of Ferrag et al. (*ICT Express* 2025, arXiv:2506.23260), under an evidence bar requiring both a static search *and* a runtime probe before marking any control absent.
- Found and remediated a missing authorization check on the agent-registry write path; fix verified by regression test (401 unauthenticated / 403 wrong-permission / 201 authorised).

Agentic-AI attacker benchmark

github.com/secslayer/finbot-bench

- Built an evaluation harness pitting local LLMs (via Ollama) as autonomous attackers against a live, deliberately-vulnerable multi-agent AI target (OWASP FinBot CTF), scored by the target's own event-driven exploit detector as ground truth — no self-judging.
- Compared three attacker architectures (single-shot / agentic loop / multi-agent planner-reflection-evolution); honest result — single-shot fails workflow execution, the multi-agent attacker produces coherent social engineering yet misses via strategic drift: attacker *architecture*, not model size, dominates.

ENGINEERING PROJECTS

Secure Guard — AI-driven incident response & SOC automation

github.com/secslayer/ai-soc

- Telemetry pipeline (Sysmon/Winlogbeat → Logstash → Elasticsearch) normalising 45K+ events; XGBoost/SVM with PCA-tuned cross-validation at ~85% accuracy; Prophet forecasting turned thresholds into concrete response playbooks.

GptGram — production-ready agentic AI platform

github.com/secslayer/Gptgram-production-ready

- AWS orchestration backend — REST APIs over distributed Celery queues for high-concurrency multi-agent LLM inference; semantic RAG on Qdrant, conversational state to S3, under Git/CI-CD.

Also built: **Ggram**, an agentic platform using graph neural networks to solve ML routing tasks; a scikit-learn SOC chatbot that cut simulated false positives by ~30% (not public).

TECHNICAL SKILLS

- **Languages & ML:** Python, C++ (incl. multithreading), SQL; CUDA (incl. cuFile) GPU programming; PyTorch, TensorFlow, JAX, scikit-learn, XGBoost, Pandas/NumPy, Databricks, Power BI; core deep-learning architectures (attention, backprop, RNN/LSTM, SVM) derived by hand.
- **AI security & agents:** jailbreak evaluation, LLM red-teaming, adversarial ML, judge/eval validity, prompt injection, activation steering/probes; multi-agent orchestration, RAG, Qdrant, LangChain.
- **Professional:** clear technical writing for non-specialists, honest reporting of negative results, teamwork on undergraduate team projects.
- **Defensive & cloud:** detection engineering, SIEM (ELK, Splunk), incident response, MITRE ATT&CK; Grafana, Prometheus; Linux, network programming; AWS (EC2, S3), Docker, Git/CI-CD, REST APIs, Agile/Scrum.

CERTIFICATIONS

- **CompTIA CySA+** (CS0-003) — [verify](#) | **Mathematics for Machine Learning**, Imperial College London / Coursera — [verify](#) | **SOC Analyst**, TryHackMe — [certificate](#)
- **In progress:** AWS Certified Generative AI Developer (AIP-C01) — prep course completed, exam not yet sat | HTB Certified Offensive AI Expert (COAE) — AI red-teamer path, in progress.

EDUCATION

University of Birmingham — MSc, Artificial Intelligence & Machine Learning

Sept 2026 – Sept 2027 (in progress)

Modules: Intelligent Robotics, Natural Language Processing, Evolutionary Computation.

TKM Institute of Technology — B.Tech, Computer Science & Engineering

CGPA 7.56/10.00

Modules: Artificial Intelligence, Data Structures, Algorithm Analysis & Design, Computer Networks.

Also: Kaggle competitor (ARC Prize 2026, ARC-AGI-3); ACM chapter participation; TryHackMe and HTB workshops on detection engineering and MITRE ATT&CK.